

Actividad 1.2 – Instalación de un servidor Apache en EC2

Alumno: Pablo **Módulo:** Despliegue de Aplicaciones Web **Práctica:** Instalación y verificación de Apache en una instancia EC2

1. Comprobación previa

Estado de la instancia:

✓	Name ↗	ID de la instancia	Estado de la i... ▼	Tipo de inst... ▼	Comprobación de	Zona de dispon... ▼	DNS de IPv4 pública ▼	Dirección IP... ▼	IP elástica ▼	Direcciones I...
✓	Unidad 1 - Pra...	i-069b43e9f58a61a36	En ejecución	t3.micro	Inicializando	us-east-1a	ec2-98-88-253-136.co...	98.88.253.136	-	-

Comando utilizado para conectar:

```
ssh -i ~/Descargas/PGG_KEY.pem ubuntu@TU_IP_PUBLICA
```

```
daw2@IF02-18:~$ ssh -i ~/Escritorio/DAW2/PGG KEY.pem ubuntu@98.88.253.136
The authenticity of host '98.88.253.136 (98.88.253.136)' can't be established.
ED25519 key fingerprint is SHA256:xEPV5PvqtfYM+42Ce18vCXDm7PvYUaG3qKojBJKV7mE.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '98.88.253.136' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1012-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep 29 15:46:39 UTC 2026

System load:  0.02           Temperature:   -273.1 C
Usage of /:   39.8% of 6.61GB Processes:    122
Memory usage: 26%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.24.242

Expanded Security Maintenance for Applications is not enabled.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Sep 29 15:44:51 2026 from 18.206.107.29
ubuntu@ip-172-31-24-242:~$ ^C
ubuntu@ip-172-31-24-242:~$
```

2. Apertura del puerto HTTP (80) en el grupo de seguridad

Describe brevemente que has accedido al grupo de seguridad y añadido la regla:

- Tipo: HTTP
- Puerto: 80
- Origen: 0.0.0.0/0

Captura:

i-069b43e9f58a61a36 (Unidad 1 - Practica pem)

Detalles

Estado y alarmas

Monitoreo

Seguridad

Redes

Almacenamiento

Etiquetas

▼ Detalles de seguridad

Rol de IAM

-

ID del propietario

153875550387

Hora de lanzamiento

Tue Sep 29 2026 17:42:15 GMT+0200 (hora de verano de Europa central)

Grupos de seguridad

sg-06f2286a8cf1c96ce (launch-wizard-2)

Editamos las reglas de entrada:

sg-06f2286a8cf1c96ce - launch-wizard-2

Acciones

Detalles

Nombre del grupo de seguridad

launch-wizard-2

ID del grupo de seguridad

sg-06f2286a8cf1c96ce

Descripción

launch-wizard-2 created 2026-09-22T15:48:46.345Z

ID de la VPC

vpc-0b9687f5d1c6f7f1e

Propietario

153875550387

Número de reglas de entrada

1 Entrada de permiso

Número de reglas de salida

1 Entrada de permiso

<

Reglas de entrada

Reglas de salida

Compartiendo

Asociaciones de VPC

Recursos relacionados

>

Reglas de entrada (1)

Administrar etiquetas

Editar reglas de entrada

Buscar

< 1 >

Name

▼

ID de la regla del gr...

▼

Versión de IP

▼

Tipo

▼

Protocolo

▼

-

sg-0a64ea0c0b4eac74e

IPv4

SSH

TCP

Editar reglas de entrada [Información](#)

Las reglas de entrada controlan el tráfico entrante que puede llegar a la instancia.

Reglas de entrada [Información](#)

ID de la regla del grupo de seguridad	Tipo Información	Protocolo Información	Intervalo de puertos Información	Origen Información	Descripción: opcional Información	
sgr-0a64ea0c0b4eac74e	SSH	TCP	22	Pe...	Q	Eliminar
-	HTTP	TCP	80	An...	Q 0.0.0.0/0	Eliminar

[Agregar regla](#)

⚠ Las reglas cuyo origen es 0.0.0.0/0 o ::/0 permiten a todas las direcciones IP acceder a la instancia. Recomendamos configurar reglas de grupo de seguridad para permitir el acceso únicamente desde direcciones IP conocidas.

[Cancelar](#) [Previsualizar los cambios](#) [Guardar reglas](#)

Diferencia entre firewall de AWS y firewall interno del sistema:

- El **grupo de seguridad** es un firewall externo gestionado por AWS, que controla el tráfico antes de llegar a la instancia.
- Un firewall interno como **ufw** controla el tráfico *dentro* del sistema operativo.

3. Instalación de Apache

Actualización de repositorios:

```
ubuntu@ip-172-31-24-242:~$ sudo apt update
```

Instalación de Apache:

```
ubuntu@ip-172-31-24-242:~$ sudo apt install apache2 -y
```

4. Verificación del servicio

Comando para ver el estado del servicio:

```

ubuntu@ip-172-31-24-242:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: v
   Active: active (running) since Tue 2026-09-29 15:56:42 UTC; 29s ago
  Invocation: 5d61e744282a4fbe81e8ad9c0fae0092
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 2344 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes>
    Tasks: 55 (limit: 627)
   Memory: 5.8M (peak: 6.1M)
      CPU: 85ms
    CGroup: /system.slice/apache2.service
            └─2344 /usr/sbin/apache2 -k start -DFOREGROUND
              └─2360 /usr/sbin/apache2 -k start -DFOREGROUND
                └─2361 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 29 15:56:41 ip-172-31-24-242 systemd[1]: Starting apache2.service - The Apa>
Sep 29 15:56:42 ip-172-31-24-242 systemd[1]: Started apache2.service - The Apac>
lines 1-17/17 (END)

```

Comando para comprobar si arranca automáticamente:

```

ubuntu@ip-172-31-24-242:~$ systemctl is-enabled apache2
enabled

```

5. Comprobación desde dentro de la instancia

Comando para hacer una petición HTTP local:

```

ubuntu@ip-172-31-24-242:~$ curl http://localhost

```

⏏ ← → ↻ http://localhost 🌐 ⚙️ VPN (



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf`. See their respective man pages for detailed information.
- The binary is called `apache2` and is managed using `systemd`, so to start/stop the service use `systemctl start apache2` and `systemctl stop apache2`, and use `systemctl status apache2` and

6. Comprobación desde fuera de la instancia

Comprobación desde navegador:



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Comprobación desde tu terminal local:

```
ubuntu@ip-172-31-24-242:~$ curl http://98.88.253.136/
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

      body, html {
        padding: 3px 3px 3px 3px;

        background-color: #D8DBE2;

        font-family: Ubuntu, Verdana, sans-serif;
        font-size: 11pt;
        text-align: center;
      }

      div.main_page {
        position: relative;
        display: table;

        width: 800px;

        margin-bottom: 3px;
        margin-left: auto;
        margin-right: auto;
        padding: 0px 0px 0px 0px;

        border-width: 2px;
```

7. Personalización básica

Directorio donde Apache sirve archivos:

```
ubuntu@ip-172-31-24-242:/var/www/html$ ls -l
total 12
-rw-r--r-- 1 root root 10672 Sep 29 15:56 index.html
ubuntu@ip-172-31-24-242:/var/www/html$
```

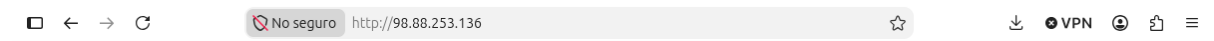
Acción realizada: Sustituir index.html por una página propia con tu nombre.

Comando utilizado:

```
ubuntu@ip-172-31-24-242:/var/www/html$ sudo nano /var/www/html/index.html
```

Antes de acceder a la página actualizada reiniciamos el servidor apache para que se apliquen los cambios utilizando el comando:

sudo systemctl restart apache2



Pablo Gonzalez Garcia

8. Localización de logs de Apache

Logs de acceso:

/var/log/apache2/access.log

```
GNU nano 8.7.1 log/apache2/access.log
::1 - - [29/Sep/2026:15:58:35 +0000] "GET / HTTP/1.1" 200 10927 "-" "curl/8.18.0"
213.96.236.200 - - [29/Sep/2026:15:59:54 +0000] "GET / HTTP/1.1" 200 3454 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:155.0) Gecko/20100101 Firefox/155.0"
213.96.236.200 - - [29/Sep/2026:15:59:54 +0000] "GET /icons/ubuntu-logo.png HTTP/1.1" 200 3607 "http://98.88.253.136/" "Mozilla/5.0 (X11; Linux x86_64; rv:155.0) Gecko/20100101 Firefox/155.0"
213.96.236.200 - - [29/Sep/2026:15:59:55 +0000] "GET /favicon.ico HTTP/1.1" 404 532 "http://98.88.253.136/" "Mozilla/5.0 (X11; Linux x86_64; rv:155.0) Gecko/20100101 Firefox/155.0"
98.88.253.136 - - [29/Sep/2026:16:00:43 +0000] "GET / HTTP/1.1" 200 10927 "-" "curl/8.18.0"
139.135.46.71 - - [29/Sep/2026:16:02:40 +0000] "GET /boaform/admin/formLogin?username=admin&psd=admin HTTP/1.0" 404 511 "-" "-"
213.96.236.200 - - [29/Sep/2026:16:08:21 +0000] "GET / HTTP/1.1" 200 331 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:155.0) Gecko/20100101 Firefox/155.0"
213.96.236.200 - - [29/Sep/2026:16:08:46 +0000] "GET / HTTP/1.1" 200 329 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:155.0) Gecko/20100101 Firefox/155.0"
```

Sirve para registrar todas las peticiones que llegan al servidor.

Logs de error:

/var/log/apache2/error.log

```
[Tue Sep 29 15:56:42.022786 2026] [mpm_event:notice] [pid 2344:tid 2344] AH00489: Apache/2.4.66 (Ubuntu) configured -- resuming normal operations
[Tue Sep 29 15:56:42.022877 2026] [core:notice] [pid 2344:tid 2344] AH00094: Command line: '/usr/sbin/apache2 -D FOREGROUND'
[Tue Sep 29 16:07:50.932299 2026] [mpm_event:notice] [pid 2344:tid 2344] AH00492: caught SIGWINCH, shutting down gracefully
[Tue Sep 29 16:07:51.035706 2026] [mpm_event:notice] [pid 8900:tid 8900] AH00489: Apache/2.4.66 (Ubuntu) configured -- resuming normal operations
[Tue Sep 29 16:07:51.035820 2026] [core:notice] [pid 8900:tid 8900] AH00094: Command line: '/usr/sbin/apache2 -D FOREGROUND'
[Tue Sep 29 16:08:14.935200 2026] [mpm_event:notice] [pid 8900:tid 8900] AH00492: caught SIGWINCH, shutting down gracefully
[Tue Sep 29 16:08:15.053478 2026] [mpm_event:notice] [pid 9924:tid 9924] AH00489: Apache/2.4.66 (Ubuntu) configured -- resuming normal operations
[Tue Sep 29 16:08:15.054246 2026] [core:notice] [pid 9924:tid 9924] AH00094: Command line: '/usr/sbin/apache2 -D FOREGROUND'
[Tue Sep 29 16:08:44.651363 2026] [mpm_event:notice] [pid 9924:tid 9924] AH00492: caught SIGWINCH, shutting down gracefully
[Tue Sep 29 16:08:44.737248 2026] [mpm_event:notice] [pid 10131:tid 10131] AH00489: Apache/2.4.66 (Ubuntu) configured -- resuming normal operations
[Tue Sep 29 16:08:44.737362 2026] [core:notice] [pid 10131:tid 10131] AH00094: Command line: '/usr/sbin/apache2 -D FOREGROUND'
```

Sirve para registrar errores del servidor y problemas de configuración.

9. Cuestiones finales

Paquetes instalados al instalar Apache

El paquete principal es **apache2**, que incluye el servidor HTTP y módulos básicos.

Usuario que ejecuta Apache

Apache se ejecuta como el usuario **www-data**. No se ejecuta como root por seguridad: si el servidor fuera comprometido, el atacante no tendría privilegios de administrador.

Diferencia entre stop, restart y reload

- **stop:** detiene el servicio.
- **restart:** detiene y vuelve a iniciar el servicio.
- **reload:** recarga la configuración sin detener el servicio.

Puerto 80 permitido pero Apache inactivo

Si el grupo de seguridad permite el puerto 80 pero Apache está detenido, no habrá respuesta. Si Apache está activo pero el puerto 80 está bloqueado, tampoco habrá acceso externo.

Qué es un virtual host

Un **virtual host** permite que un mismo servidor Apache atienda varios sitios web diferentes usando la misma instancia.

Ventajas e inconvenientes de exponer Apache directamente a Internet

Ventajas:

- Simplicidad

- Menos componentes
- Fácil de configurar

Inconvenientes:

- Menor seguridad
- No hay balanceo de carga
- No hay protección frente a tráfico malicioso
- No escala bien en producción

10. Fuentes consultadas

- Documentación oficial de Apache
- Documentación de AWS EC2
- Apuntes del módulo